

Final Endeavor for the Generalized t -SNE Problem

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1 Progress So Far and Setup

OK this is it. We have reached this step and the end is within our grasp. The stake is high and I can literally feel the adrenaline coursing my vein, along with my fear. So far we have finished two crucial steps. First, we have finished the proof of convergence for the generalized output kernel. Denote $\mathbf{Ker}_\mathbf{O}$ the output kernel and $\mathbf{Ker}_\mathbf{I}$ the input kernel. The main theorems in [1] will hold if the following is true:

Condition 1 (Output Kernel Conditions). *The output kernel $\mathbf{Ker}_\mathbf{O}(y, y') = k(\|y - y'\|)$ depends only on the distance between y and y' and that:*

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded;
2. g satisfies

$$\iint_{(0, \infty)^2} g(y, y') dy dy' < \infty; \quad (1.1)$$

3. k has bounded derivative;
4. $k'(0) = 0$.

Moreover, the lemmas that deal with the output does not interfere with those related to the input kernel. The initial exploration we did around the output kernel not only provides us a good opportunity to understand the problem and the literature, but also serves as an integral part of the final paper. If we are able to tackle the lemmas related to the input kernel we will be able to join the results seamlessly.

On the other hand, we have established the form of the generalized input kernel, set a basis for us to work on, and prove the existence of $\sigma_{\rho, \mu}^*(x)$, the unique solution to $F_{\rho, \mu}(x, \sigma) = 0$. For this to happen, the generalized input kernel should satisfy the following conditions:

Condition 2 (Initial Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_\mathbf{I}(x_i, x_j, \sigma_i) = \exp(-w(\sigma_i \|x_j - x_i\|^\theta))$. Here, σ_i is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. (We can replace σ_i with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$ in the continuous version of the kernel.) w satisfies:*

1. $w'(t) > 0$ for all $t \geq 0$ with $\lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following two integrals converge:

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty, \quad \int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (1.2)$$

Since the probability masses are all normalized, we can assume without loss of generality that $w(0) = 0$. Note that condition 2 is only enough to prove the existence and uniqueness of σ^* . We may add more assumptions later on if necessary. Before we march towards the final battle, let us remind ourselves some important definitions and notations, as well as some of the lemmas from [1] that we can borrow. Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 1. *Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that*

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (1.3)$$

Lemma 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that*

$$\mathbb{P}\left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t\right) \leq Ce^{-c_1 t^2}. \quad (1.4)$$

Lemma 3. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\sigma \|x - x'\| \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma^{-1} \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (1.5)$$

Finally, let us also recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (1.6)$$

The two terms in the formula of F are the pillars of this manuscript. Their convergence will lead us to prove the convergence of F , and most importantly, the uniform convergence of $\sigma(x)$. **[Yi: As for the question Tuca asked me during our discussion, yes we can define $\sigma(x)$ to be the unique solution to $F(x, \sigma) = 3$ or whatever number we peek as long as the number is greatly than zero (to ensure the existence of uniqueness of the solution). If we move the constant to the left it is equivalent to just pick a different perplexity. The convergence of F works for any σ and that is the key. As long as control the entropy, we are expecting convergence as long as we set up the continuous version to have the same perplexity.]**

2 The Two Most Important Lemmas

The biggest obtascles left are to proof two lemmas similar to lemma 6 and lemma 7 in [1]. These two lemmas will give us the convergence of F and thus the convergence of $\sigma(x)$. We

will try to tackle this part using similar methods from the original paper. Write

$$\begin{aligned} f_\sigma(x, x') &= \exp(-w(\sigma \|x - x'\|^\theta)), \\ g_\sigma(x, x') &= w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)). \end{aligned} \quad (2.1)$$

Note that for any real numbers A, B, C, D , we have

$$\left| \frac{A}{B} - \frac{C}{D} \right| = \left| \frac{AD - BC}{BD} \right| = \left| \frac{D(A - C) + C(D - B)}{BD} \right| \leq \left| \frac{A - C}{B} \right| + \left| \frac{C(D - B)}{BD} \right|. \quad (2.2)$$

This means

$$\begin{aligned} \left| \frac{\int g_\sigma(x, x') d\mu_n(x')}{\int f_\sigma(x, x') d\mu_n(x')} - \frac{\int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu(x')} \right| &\leq \left| \frac{\int g_\sigma(x, x') d\mu_n(x') - \int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu_n(x')} \right| \\ &+ \left| \frac{\int g_\sigma(x, x') d\mu(x') (\int f_\sigma(x, x') d\mu_n(x') - \int f_\sigma(x, x') d\mu(x'))}{\int f_\sigma(x, x') d\mu(x') \int f_\sigma(x, x') d\mu_n(x')} \right|. \end{aligned} \quad (2.3)$$

Let $\alpha > 0$. Define

$$E'_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\left(\alpha + \frac{1}{c_1}\right) \sqrt{\log n} + \sqrt{d}} \right\}. \quad (2.4)$$

Setting $t = \sqrt{\alpha \log n}$ in lemma 2 we have $\mathbb{P}(E'_1) \leq Cn^{-c_1\alpha}$. To absorb the $\sqrt{d}$ term in E'_1 , we can find $\epsilon_n \rightarrow 0$ such that

$$E_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\left(\alpha + \frac{1}{c_1} + \epsilon_n\right) \sqrt{\log n}} \right\} \quad (2.5)$$

with $\mathbb{P}(E_1) \leq Cn^{-c_1\alpha}$. On the other hand, on E_1^c for all i, j ,

$$\|X_i - X_j\| \leq 2\sqrt{\left(\alpha + \frac{1}{c_1} + \epsilon_n\right) \sqrt{\log n}} =: 2\gamma\sqrt{\log n}. \quad (2.6)$$

This means for $\sigma^{-1} \in [\delta^\theta, n^{\theta/2}]$ on E_1^c we have

$$\begin{aligned} \int f_\sigma(X_i, x') d\mu_n(x') &= \int \exp(-w(\sigma \|X_i - x'\|^\theta)) d\mu_n(x') \\ &\geq \exp\left(-w\left(\left(\frac{2\gamma\sqrt{\log n}}{\delta}\right)^\theta\right)\right) =: T. \end{aligned} \quad (2.7)$$

[Yi: This is where the interesting part begins. The proofs in the original paper shows us that we do not want our kernel to decay too fast otherwise the bounds cannot be controlled.] In a same way, by lemma 1, we get similar bounds

$$\int f_\sigma(X_i, x') d\mu(x') \geq (1 - Cm^{-c_1\alpha})T \geq \frac{1}{2}T. \quad (2.8)$$

By lemma 3 with $h = f_\sigma$ and $h = g_\sigma$, there exists E_2 with $\mathbb{P}(E_2) \leq 2e^{-c\epsilon^2 n}$ such that on E_2^c ,

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| \\ + \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \int g_\sigma(X_i, x') d\mu_n(x') - \int g_\sigma(X_i, x') d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \end{aligned} \quad (2.9)$$

Therefore on $E_1^c \cap E_2^c$, combining (2.3), (2.7), (2.8), and (2.9) gives us

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(x, x') d\mu_n(x')}{\int f_\sigma(x, x') d\mu_n(x')} - \frac{\int g_\sigma(x, x') d\mu(x')}{\int f_\sigma(x, x') d\mu(x')} \right| \leq C\epsilon T^{-1} + O\left(\frac{\sqrt{\log n}}{n^{1/2}T}\right). \quad (2.10)$$

Again, we have reduced the core parts of this problem into a calculus exercise. We want RHS goes to zero as $n \rightarrow \infty$. Towards this end we need

$$\epsilon T^{-1} \rightarrow 0, \quad T^{-1} \cdot \frac{\sqrt{\log n}}{n^{1/2}} \rightarrow 0. \quad (2.11)$$

On the other hand, since $\mathbb{P}(E_1 \cap E_2) \leq Cn^{-c_1\alpha} + 2e^{-c\epsilon^2 n}$, we also need

$$c_1\alpha > 1, \quad \epsilon^2 n \rightarrow \infty. \quad (2.12)$$

Let $\epsilon = n^{-\eta}$ and by (2.11) we have $0 < \eta < 1/2$. Then, if we have $n^{-\eta}T^{-1} \rightarrow 0$ for some $\eta \in (0, 1/2)$, then we must have

$$T^{-1} \cdot \frac{\sqrt{\log n}}{n^{1/2}} \leq \frac{\sqrt{\log n}}{n^{1/2-\eta}} \rightarrow 0. \quad (2.13)$$

This means the first inequality in (2.11) implies the second. The requirements in (2.11) and (2.12) are equivalent to

$$n^{-\eta}T^{-1} \rightarrow 0, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (2.14)$$

i.e.,

$$n^{-\eta} \exp\left(w \left(\left(\frac{2\gamma\sqrt{\log n}}{\delta}\right)^\theta\right)\right) \rightarrow 0, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (2.15)$$

Let $\alpha = 1/c_1 + \epsilon_1$ and $\eta = 1/2 - \epsilon_2$. Then the above requirements are equivalent to

$$\lim_{n \rightarrow \infty} n^{-1/2+\epsilon_2} \exp\left(w \left(\left(\sqrt{\frac{8}{(c_1 + \epsilon_1)\delta^2}} \sqrt{\log n}\right)^\theta\right)\right) = 0. \quad (2.16)$$

As ϵ_1 and ϵ_2 can be arbitrarily small, the above limit can be further simplified to

$$\lim_{n \rightarrow \infty} n^{-1/2} \exp\left(w \left(\left(\sqrt{\frac{8}{c_1\delta^2}} \sqrt{\log n}\right)^\theta\right)\right) = 0. \quad (2.17)$$

Like [1], the quantity $c_1\delta^2$ shows up again as our slider. It is just less explicit. Also, we need to choose w that does not increase too fast that the above limit will fail. To put it in a more formal way, consider the set below

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \mid \lim_{n \rightarrow \infty} n^{-1/2} \exp\left(w \left(\left(\tilde{C}\sqrt{\log n}\right)^\theta\right)\right) = 0 \right\}. \quad (2.18)$$

Obviously, $0 \in \mathcal{K}_w$ so it is nonempty. If there exists $\tilde{C}' \in \mathcal{K}_w$ such that $\tilde{C}' > 0$. Then for any $0 \leq \tilde{C}'' < \tilde{C}'$, we have $\tilde{C}'' \in \mathcal{K}_w$ by monotonicity of w . Inspired by this fact, we can require

$$\sup \mathcal{K}_w > 0. \quad (2.19)$$

Then for any $\tilde{C} < \sup \mathcal{K}_w$, $\tilde{C} \in \mathcal{K}_w$. Back to (2.17), we simply need to let

$$c_1\delta^2 > \frac{8}{(\sup \mathcal{K}_w)^2}. \quad (2.20)$$

Now we are ready to formally state the two MOST important lemmas for this entire project.

Condition 3. Define

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \left| \lim_{n \rightarrow \infty} n^{-1/2} \exp \left(w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right. \right\}. \quad (2.21)$$

Then w satisfies that

$$\sup \mathcal{K}_w > 0. \quad (2.22)$$

Lemma 4. Let c_1 be the decay constant of the sub-Gaussian distribution μ , and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Then almost surely in $X_1, \dots, X_n$,

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \\ = O \left(n^{-1/2} \exp \left(w \left(\left(\sqrt{\frac{8}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right) \right). \end{aligned} \quad (2.23)$$

Using the elementary inequality $\log(x) \leq x - 1$, we can prove the next lemma in a similar way:

Lemma 5. Let c_1 be the decay constant of the sub-Gaussian distribution μ , and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Then almost surely in $X_1, \dots, X_n$,

$$\begin{aligned} \sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| \\ = O \left(n^{-1/2} \exp \left(w \left(\left(\sqrt{\frac{4}{c_1 \delta^2}} \sqrt{\log n} \right)^\theta \right) \right) \right). \end{aligned} \quad (2.24)$$

Then using these two lemmas we have the following two theorems:

Proposition 1. Let μ be sub-Gaussian with decay constant c_1 from lemma 1 and $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . $\mu_n = (1/n) \sum_i \delta_{X_i}$ is the empirical measure. Set $\delta > 0$ to be a constant such that $c_1 \delta^2 > 8/(\sup \mathcal{K}_w)^2$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma^{-1} \in [\delta^\theta, n^{\theta/2}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = o(1). \quad (2.25)$$

Since in [1] the convergence of σ^* does not use the polynomial decay in the original paper, we have the following milestone proposition as well:

Proposition 2. Suppose μ has compact support and C_1 density. Then almost surely, as $n \rightarrow \infty$,

$$\sup_{i=1, \dots, n} |\sigma_{\mu_n}^*(X_i) - \sigma_\mu^*(X_i)| \rightarrow 0. \quad (2.26)$$

3 The Rest of the Proof

It is not over yet, as in [1] some of the remaining lemmas require polynomial decay and we need to thread carefully and tweak condition 3 to make sure the rest of the proof goes through, along with the generalized output kernels.

4 Wrap Up and Conclusion

References

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